Homework

Robotics (B-KUL-H02A4A)

2026

Instructions Any sketch or figure in your answer can be a photo or scan of a readable figure. If plots or numerical results are requested, provided the (python) **code** that generate these results in the form of a *runnable* jupyter notebook (with a name indicating the homework and the number of the question). Copy the numerical result or plot in your answer document.

1. Sample randomly 10 000 times from the distribution with probability density function:

$$p(x) = 0.5\mathcal{N}(x| - 1.0, 1) + 0.5\mathcal{N}(x|1.0, 0) \tag{1}$$

   where the function $\mathcal{N}(x|\mu, P)$ corresponds to the probability density function of a normal distribution with mean μ and variance P. Does the distribution need to be normalized for your program to work? Plot a histogram of the samples.

   a) Sample randomly 10 000 times from the distribution with given probability density function and plot a histogram of the samples. Give the mean and standard deviation up to 5 digits precision.

   b) Assume we do a measurement with model $z = 2x + \varepsilon$, with $\varepsilon = \mathcal{N}(0, 0.3)$ (i.e. normal distributed with variance 0.3). The measurement returns 0.5. Perform the measurement and plot a histogram of the samples Give the mean and standard deviation up to 5 digits.

   c) How is the resulting distribution called?

2. Given a 6-dof robot in joint position $\boldsymbol{q}$ with a 6×6 Jacobian $\boldsymbol{J}(\boldsymbol{q})$, and a plane through the end effector. The plane is further described by two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ inside the plane. These vectors are not parallel and are not orthogonal to each other. The joint velocities $\dot{\boldsymbol{q}}$ are limited to $\dot{\boldsymbol{q}} \leq 1$

   a) What is the shape of the set of all reachable velocities inside the plane?

   b) what is the magnitude of the maximum velocity that the robot can move inside the plane? Do not use a numerical optimization, but only rely on linear algebra operations.

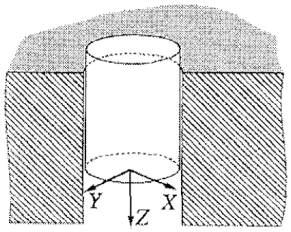
3. $\mu-$parameterization of inertia

   (a) How do you transform mass parameters of link $\boldsymbol{\mu}^a$ expressed in a frame $\{a\}$ to mass parameters $\boldsymbol{\mu}^b$ for the same link expressed in a different frame $\{b\}$?

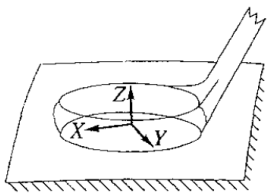
   (b) How can you verify wheter a mass parameterization $\boldsymbol{\mu}$ is physically realizable?

   (c) I have a body a with inertia parameters $\boldsymbol{\mu}^a$, and body b with inertia parameters $\boldsymbol{\mu}^a$ that are rigidly linked, what are the $\mu-$parameters of the joint body ab?

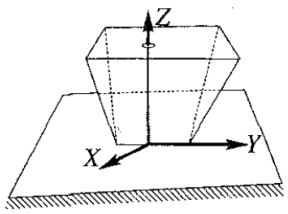
4. For the three cases described below indicate i) all of the natural constraints ii) all of the artificial constraints iii) the specification of a task according to the task frame formalism. For all cases, it is assumed that contact is frictionless and infinitely stiff. In your answers, consider forces and moments acting **on** the object, expressed in the indicated frame. You can indicate arbitrary values with a symbol, while indicating if there are positive , negative or arbitrary.



(a) Peg insertion



(b) Flat object on plane



(c) Wedge on a plane

   a) Give the natural and artificial constraints and specification in the task frame formalism specification for peg insertion (fig. 1a). The peg is cylindrical and is inserted in a round hole.

   b) Give the natural and artificial constraints and specification in the task frame formalism specification for the motion of a flat object on a plane (fig. 1b). The bottom of the object is planar.

   c) Give the natural and artificial constraints and specification in the task frame formalism specification for a wedge on a plane (fig. 1c). The bottom edge of the wedge is a straight line. The surface is planar.

   d) Under the assumptions of frictionless and infinitely stiff contact, and in an aribrary frame attached to the object, what can you say about the relationship between screw twist and wrench (both expressed in that arbitrary frame) for the cases described above?